

EQUATIONS

CONVERSIONS

1 L = 1 dm³
 1 mL = 1 cm³
 1 g = 1 mL of water at 4°C
 1 inch = 2.54 cm
 1 foot = 12 inches
 1 metre = 39.37 in
 1 teaspoon (tsp) = 5 mL
 1 tablespoon (tbsp) = 15 mL
 1 fluidounce (fl. oz.) = 29.57 mL
 1 pint (16 fl. oz.) = 473 mL
 1 quart (32 fl. oz.) = 946 mL
 1 US gallon (128 fl. oz.) = 3785 mL
 1 pound (lb) = 454 g
 1 kilogram = 2.2 lbs

MEDICAL LATIN ABBREVIATIONS

BID: twice daily TID: three times daily
 QID: four times daily GTT: drop PO:
 by mouth/orally IV: intravenously OS:
 left eye OD: right eye OU: both eyes
 AS: left ear AD: right ear AU: both ears

TERMS:

Median effective dose (ED50)
 Amount that produces desired effect in 50% of tested subjects
 Median toxic dose (TD50)
 Amount that produces toxic effects in 50% subjects
 Minimum effective concentration (MEC)
 Minimum blood serum concentration expected to give the desired effect
 Maximum tolerated dose (MTC)
 Maximum blood serum concentration before undesired effects occur
 Therapeutic window (MTC-MEC)
 Serum concentration where a medication is effective but not toxic

BODY MASS INDEX (BMI)

$$BMI = \frac{\text{weight (kg)}}{[\text{height (m)}]^2}$$

BMI Range	Status
< 18.5	Underweight
18.5 – 24.9	Normal
25 – 29.9	Overweight
> 30	Obese

Millimoles (mmol)

Mole = measure of # of particles in a substance
 1 mol = 6.02 x 10²³ particles

Milliosmoles (mOsmols)

of mmols multiplied by the # of ions or parts that a substance can dissociate into when placed in solution

Buffer solutions

Ka: dissociation constant of an acid
 Kb: dissociation constant of a base

$$K_a = \frac{[H^+][A^-]}{[HA]} \quad K_b = \frac{[HB^+][OH^-]}{[B]}$$

$$pK_a = -\log(K_a)$$

$$pK_b = -\log(K_b)$$

$$pH + pOH = pK_w = 14$$

$$pH = -\log[H^+]$$

$$pH = pK_a - \log\left(\frac{[A^-]}{[HA]}\right)$$

$$pOH = pK_b + \log\left(\frac{[HB^+]}{[B]}\right)$$

Isotonicity

$$\Delta T_f (\text{bodily fluid}) = -0.52^\circ\text{C}$$

$$\Delta T_f (\text{bodily fluid}) = \Delta T_f (\text{drug solution}) = \text{isotonic}$$

10 ⁻⁶	10 ⁻³	10 ⁻²	10 ⁻¹	10 ⁰	10 ¹	10 ²	10 ³	10 ⁶
micrometre (µm)	millimetre (mm)	centimetre (cm)	decimetre (dm)	metre (m)	decametre (dam)	hectometre (hm)	kilometre (km)	megametre (Mm)
÷ 1000000	÷ 1000	÷ 100	÷ 10	-	x 10	x 100	x 1000	x 1000000

OPIOID CONVERSIONS

Note: reduce new dose by 25-50% for incomplete cross-tolerance

Drug	Oral dose (mg)	IV/IM/SC dose (mg)
Morphine	30	10
Oxycodone	20	-
Hydromorphone	6	1.5
Codeine	200	-
Tramadol	150	-
Methadone	Morphine dose equivalence not reliably established	

IDEAL BODY WEIGHT (IBW)

Women: 45.5 kg + 2.3kg x [# of inches above 5 feet in height]

Men: 50 kg + 2.3kg x [# of inches above 5 feet in height]

ADJUSTED BODY WEIGHT (ADJBW)

AdjBW = IBW + 0.4 x (Actual Body Weight – Ideal Body Weight)

CREATININE CLEARANCE (CrCl)

If Actual Body Weight (ABW) < Ideal Body Weight (IBW), use ABW

If IBW < ABW < 130% of IBW, use IBW

If 130% of IBW < Actual Body Weight, use Adjusted Body Weight (AdjBW)

$$CrCl \left(\frac{\text{mL}}{\text{min}} \right) = \frac{(140 - \text{age}) \times 1.2 \times \text{weight (kg)}}{\text{serum creatinine} \left(\frac{\mu\text{mol}}{\text{L}} \right)} \times 0.85 \text{ (if female)}$$

$$CrCl \left(\frac{\text{mL}}{\text{min}} \right) = \frac{(140 - \text{age}) \times \text{weight (kg)}}{72 \times \text{serum creatinine} \left(\frac{\text{mg}}{\text{dL}} \right)} \times 0.85 \text{ (if female)}$$

$$C_i V_i = C_f V_f$$

C: concentration V: volume I: initial F: final

$$\frac{(A) \text{ Amount of drug in dilution}}{(B) \text{ dilution volume}} = \frac{(C) \text{ Amount of drug in aliquot}}{(D) \text{ aliquot volume}}$$

MILLIEQUIVALENTS (mEq)

chemical units to express the concentration of electrolytes in solution

$$mEq = \frac{\text{mass of ion (mg)} \times \text{valence of ion}}{\text{molecular mass}}$$

OSMOLARITY (mOsmol/L)

Measure of osmotic concentration in a solution

$$\frac{\text{mOsmol}}{\text{L}} = \frac{\text{concentration (mg/L)}}{\text{molecular weight (mg/mmol)}} \times \text{number of species}$$

INFUSION FLOW RATE

$$\text{Flow Rate (drops/min)} = \frac{\text{volume (mL)} \times \text{drip set (drops/mL)}}{\text{time (minutes)}}$$

HARRIS-BENEDICT EQUATIONS

Used to estimate basal energy expenditure (BEE)

$$\text{BEE (males)} = 66.67 + [13.75 \times \text{weight (kg)}] + [5 \times \text{height (cm)}] - [6.76 \times \text{age (years)}]$$

$$\text{BEE (females)} = 655.1 + [9.56 \times \text{weight (kg)}] + [1.86 \times \text{height (cm)}] - [4.68 \times \text{age (years)}]$$

TOTAL DAILY EXPENDITURE (TDE)

Accounts for activity and stress factors

$$\text{TDE (male or female)} = \text{BEE} \times \text{activity factors} \times \text{stress factors}$$

Activity Factors	
Confined to bed	1.2
Ambulatory	1.3
Stress Factors	
Surgery	1.2
Trauma	1.3-1.5
Infection	1.4-1.6
Burns	1.5-2.1

DOSING BASED ON BODY WEIGHT AND SURFACE AREA

$$\text{patient's dose (mg)} = \text{patient's weight (kg)} \times \frac{\text{drug dose (mg)}}{1 \text{ kg}}$$

PEDIATRIC ADJUSTMENT OF DOSES

Note: in practice, these dosing rules are only used if a pediatric dosing reference is not available

$$\text{Young's Rule (age-based): pediatric dose} = \text{dose for average adult} \times \frac{\text{age (years)}}{\text{age (years)} + 12}$$

$$\text{Clark's Rule (mass-based): pediatric dose} = \text{dose for average adult} \times \frac{\text{mass of child (lbs)}}{150}$$

SPECIFIC GRAVITY

A ratio that expresses how many times heavier (or lighter) a liquid is than water. Specific gravity of water is 1. Density of water is 1 g/mL.

$$\text{specific gravity} = \frac{\text{weight of substance}}{\text{weight of equal volume of water}}$$

% volume per volume (v/v) is expressed as mL per 100 mL of solution

% weight per volume (w/v) is expressed as grams per 100 mL of solution

% weight per weight (w/w) is expressed as grams of solute per 100 grams of preparation

PARTS PER BILLION (PPB) & PARTS PER MILLION (PPM)

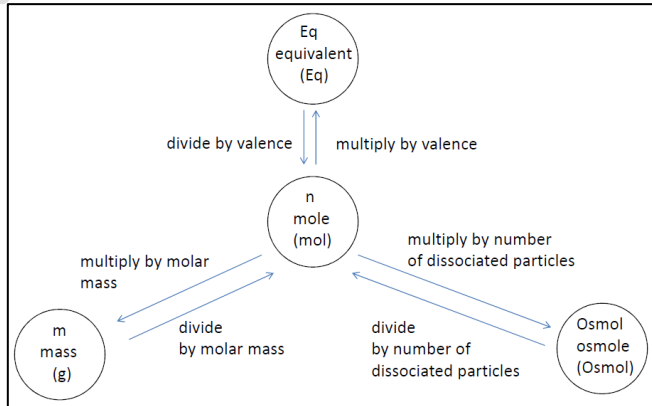
$$1 \text{ ppm} = 1 \text{ mg/kg} = 1 \text{ mg/L} = 1 \mu\text{g/g} = 1 \mu\text{g/mL}$$

$$1 \text{ ppb} = 1 \mu\text{g/kg} = 1 \mu\text{g/L} = 1 \text{ ng/g} = 1 \text{ ng/mL}$$

MINIMUM WEIGHABLE QUANTITY (MWQ)

The lowest weight that can be accurately measured

$$\text{Sensitivity} = \text{Maximum Acceptable Error (5\%)} \times \text{MWQ}$$



TOTAL PARENTAL NUTRITION

Parameter	Fluid Requirement														
Neonates	120-180 mL/kg per day														
1-12 months	150 mL/kg (max 200mL)														
< 10 kg	100 mL/kg per day														
10-20 kg	1000mL + 50 mL/kg over 10kg														
>20kg	1500mL + 20 mL/kg over 20kg														
Body surface area (BSA)	1500 mL/m ² per day														
Caloric requirement	1.2 mL/kcal per day														
Parameter	Caloric Requirements														
Carbohydrates	3.4 kcal/gram														
Protein	4 kcal/gram														
<table border="1"> <tr> <th>Patient type</th><th>Amount of protein/day</th></tr> <tr> <td>Unstressed</td><td>0.8 g/kg</td></tr> <tr> <td>Mildly stressed</td><td>0.8-1 g/kg</td></tr> <tr> <td>Moderated stressed</td><td>1.1-1.5 g/kg</td></tr> <tr> <td>Severely stressed</td><td>1.5-2 g/kg</td></tr> <tr> <td>Renal dialysis</td><td>1.2 g/kg</td></tr> <tr> <td>Severely burned</td><td>3 g/kg</td></tr> </table>	Patient type	Amount of protein/day	Unstressed	0.8 g/kg	Mildly stressed	0.8-1 g/kg	Moderated stressed	1.1-1.5 g/kg	Severely stressed	1.5-2 g/kg	Renal dialysis	1.2 g/kg	Severely burned	3 g/kg	
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Renal dialysis	1.2 g/kg														
Severely burned	3 g/kg														
Fat	10% lipid emulsion provides 11 kcal/gram														
Fiber	14 g for each 1000 calories														

EQUATIONS

EXPONENTS & LOGARITHMS

X^A, X = base, A = exponent
If X^b = Y, then log_xY = b

$$X^0 = 1, X^1 = X$$
$$X^{-a} = \frac{1}{X^a}$$
$$X^a * X^b = X^{a+b}$$
$$\frac{X^a}{X^b} = X^{a-b}$$
$$X^{a^b} = X^{a^{*b}}$$
$$(X \cdot Y)^a = X^a \cdot Y^a$$

$$\log_x a^x = x$$
$$\log(XY) = \log X + \log Y$$
$$\log \frac{X}{Y} = \log X - \log Y$$
$$\log(X^Y) = Y \log X$$
$$\log_a X = \frac{\log_b X}{\log_b a}$$
$$\log(X+Y) \neq \log X + \log Y$$

pH & pKa RELATIONSHIP

- pH – measure of [H⁺]
- pOH – measure of [OH⁻]
- Low pH = acidity
- pH of 7 = neutral
- High pH = alkalinity
- Ka = acid dissociation constant
- Kb = base dissociation constant

ONE COMPARTMENT MODEL

$$C_t = C_0 \cdot e^{-k_{el} \cdot t}$$
$$\ln(C_t) = \ln(C_0) - k_{el} \cdot t$$

TWO COMPARTMENT MODEL

$$C_t = C_1 \cdot e^{-a \cdot t} + C_2 \cdot e^{-b \cdot t}$$

LOADING DOSE

Can be used if drug has a long half-life or therapeutic concentrations are needed immediately

$$D_L = \frac{C_{ps} \cdot V_d}{S \cdot F} = \frac{D_M}{1 - e^{-k \cdot \tau}}$$

C_{ps}: steady state plasma concentration, S: salt factor, F: bioavailability factor, V_d: volume of distribution, D_L: loading dose, D_M: maintenance dose

FLUCTUATION

Difference between a peak and a trough during a single dosing interval

$$\frac{C_{pmax,ss}}{C_{pmin,ss}} = \frac{1}{e^{-k \cdot \tau}}$$

ACCUMULATION RATIO (r)

Increase that occurs in drug concentration with each additional dose

$$r = \frac{C_{ps}}{C_{p1}} = \frac{1}{1 - e^{-k \cdot \tau}}$$

RADIOACTIVE DECAY

$$\lambda = \frac{\ln(2)}{t_{1/2}}$$
$$A_t = A_0 \times e^{-\lambda t}$$

λ: decay constant, A_t: radioactivity at time t, A₀: radioactivity at t = 0

PEAK & TROUGH CALCULATIONS

C: concentration, R₀: infusion rate, V_d: distribution volume, Ke: elimination constant, t: infusion time, T: dosing interval

$$C \text{ (peak)} = \frac{R_0}{V_d K_e} \left(\frac{1 - e^{-K_{et}}}{1 - e^{-K_e T}} \right)$$

$$C \text{ (trough)} = C_{peak} \cdot e^{-K_e(T-t)}$$

PERCENT IONIZATION

Amount of acid or base that dissociates into its ions at a specific concentration. Weak acid/bases partially ionize. Strong acid/bases fully dissociate into ions.

% **ionization** for weak acid = 100/ [1 + antilog (pK_a – pH)]
% **ionization** for weak base = 100/ [1 + antilog (pH – pK_a)]
K_a: equilibrium constant of an acid
pK_a scale is defined by: pK_a = - log K_a

ELIMINATION RATE CONSTANT (k) & HALF LIFE (t_{1/2})

Measuring the speed of drug elimination. Reciprocal forms of each other. Large k = short half life. Low k = long half life.

$$k = \frac{Cl}{V_d} = \frac{\ln(C_{p1}) - \ln(C_{p2})}{t_2 - t_1} \qquad t_{1/2} = \frac{0.693}{k}$$

k: elimination rate constant, Cl: drug clearance, V_d: volume of distribution, t_{1/2}: half life

HEPATIC CLEARANCE

Volume of fluid completely cleared of drug by the liter per unit time

$$Cl_{bh} = Q \cdot E$$

Cl_{bh}: hepatic blood clearance, Q: hepatic blood flow, E: hepatic extraction ratio

WELL-STIRRED MODEL

Rate of metabolism is equal to the difference between drug presentation to the liver and the rate of drug exit from the liver

$$Cl_h = \frac{Q \cdot f_u \cdot Cl_{int}}{Q + f_u \cdot Cl_{int}}$$

Cl_h: hepatic clearance, Q: hepatic blood flow, f_u: fraction of drug unbound in the blood, Cl_{int}: intrinsic clearance

IV BOLUS PLASMA CONCENTRATION

Rate of drug over time at the start is equal to the difference between the rate of inputs and rate of outputs

$$C_p = \frac{F \cdot D}{V_d} \cdot e^{-k \cdot t} = C_{p0} \cdot e^{-k \cdot t}$$

C_p: plasma concentration, V_d: volume of distribution, F: bioavailability factor, D: dose administered, C_{p0}: plasma concentration at time zero, k: elimination rate constant

NOYES-WHITNEY EQUATION

Rate of drug dissolution

$$\frac{dC}{dt} = \frac{D \cdot S}{V \cdot h} \cdot (C_s - C)$$

C: concentration in gastrointestinal fluid, t: time, S: surface of the drug, H: thickness of the diffusion layer surrounding the solid, D: diffusion coefficient, V: volume of fluid, C_s: solubility of the drug

INTRAVENOUS INFUSION BASIC EQUATION

Rate of change of the amount of drug in the body is equal to the difference between the rate of inputs and rate of outputs

$$C_p = \frac{F \cdot R_0}{Cl} \cdot (1 - e^{-k \cdot t})$$
$$C_{p,ss} = \frac{F \cdot R_0}{Cl}$$

C_p: plasma concentration, F: bioavailability factor, Cl: drug clearance, R₀: infusion rate, k: elimination rate constant, t: time, C_{pss}: steady-state plasma concentration

PHENYTOIN

Displays non-linear PK. Narrow therapeutic range. Primarily eliminated by saturable metabolism (described by Michaelis-Menten equation)

The following equation can be used for phenytoin and other drugs that have similar saturable elimination:

$$\frac{F \cdot D}{\tau} = \frac{V_{max} \cdot C_{pss}}{K_m + C_{pss}}$$

Phenytoin steady-state concentration:

$$C_{pss} = \frac{K_m \cdot F \cdot Ra}{V_{max} - F \cdot Ra}$$

Phenytoin maximum speed:

$$V_{max} = \frac{(K_m + C_{pss}) \cdot F \cdot Ra}{C_{pss}}$$

V_{max}: maximum metabolism speed achieved at enzyme saturation, K_m: Michaelis-Menten constant, C_{pss}: steady-state plasma concentration, F: bioavailability, D: dose, τ: dosing interval, Ra: rate of drug administration

RATE OF DRUG ABSORPTION

It is controlled by the SLOWEST STEP!

- When **dissolution rate controls absorption**, the **Noyes-Whitney equation describes the rate**. The drug concentration in gastrointestinal fluid will be much smaller than the drug solubility (C_s >> C)
- When **membrane penetration controls absorption**, the **rate is approximated by Fick's law**
- Dissolution and membrane penetration approximate a first order process (A_b: amount of drug in the body, t: time, A_{GI}: amount of drug in the GI tract, k_a: rate constant for absorption)

$$\frac{dA_b}{dt} = A_{GI} \cdot k_a$$

VOLUME OF DISTRIBUTION (Vd)

Primary distribution parameter for a drug. NOT a physiological volume. V_d reflects how a drug will diffuse throughout the body.

$$V_d = \frac{A_b}{C_p}$$

A_b: amount of drug in the body at a specific time (mg), C_p: plasma concentration at the same time (mg/L)

BIOAVAILABILITY

$$F = \frac{AUC_{oral} \times Dose_{IV}}{AUC_{IV} \times Dose_{oral}}$$

AUC: area under the curve

CLEARANCE

The efficiency in which the liver/kidney can extract drug from plasma and eliminate it. Total clearance = renal + hepatic clearance

$$Cl = \frac{\text{rate of elimination}}{C_p}$$

Cl: clearance, C_p: plasma concentration

CLEARANCE USING AUC

$$Cl = \frac{D_{iv}}{AUC(0 \text{ to } \infty)} = \frac{F \cdot D}{AUC(0 \text{ to } \infty)}$$

Cl: drug clearance, Div: IV dosage, F: bioavailability, D: oral dose, AUC (0 to ∞): area under the curve from zero to infinity

RENAL CLEARANCE

Major site of drug elimination. Average clearance is 120 mL/min

$$Cl \text{ (renal)} = \frac{\text{filtration} + \text{secretion} - \text{absorption}}{C_p}$$

ORAL DOSE PLASMA CONCENTRATION

Rate of drug over time at the start is equal to the difference between the rate of inputs and rate of outputs

$$C_p = \frac{F \cdot D \cdot ka}{V_d \cdot (ka - k)} \cdot (e^{-k \cdot t} - e^{-ka \cdot t})$$

C_p: plasma concentration, V_d: volume of distribution, k: elimination rate constant, F: bioavailability, D: dose, ka: absorption rate constant, t: time

STEADY STATE BASIC EQUATION

$$C_{pss,n} = \frac{F \cdot D}{V_d \cdot (1 - e^{-k \cdot \tau})} \cdot e^{-k \cdot t}$$
$$C_{pmax,ss} = \frac{F \cdot D}{V_d \cdot (1 - e^{-k \cdot \tau})} \qquad C_{pmin,ss} = C_{pmax,ss} \cdot e^{-k \cdot \tau}$$

C_{pss}: plasma concentration, C_{pmax,ss}: maximum plasma concentration at steady state, C_{pmin,ss}: minimum plasma concentration at steady state, k: elimination rate constant, τ: dosing interval, F: bioavailability factor, V_d: volume of distribution, D: dose, t: time

AVERAGE PLASMA CONCENTRATION

$$C_{pav,ss} = \frac{AUC}{\tau} = \frac{F \cdot D}{Cl \cdot \tau} = \frac{F \cdot Ra}{Cl}$$

C_{pav,ss}: average plasma concentration at steady state, τ: dosing interval, Ra: average rate of drug administration over the dosing interval, F: bioavailability factor, Cl: clearance, D: dose

PHARMACOKINETIC PARAMETERS

Symbol	Common Units
C _{max} (max. plasma concentration)	ng/mL, ug/L, uM
T _{max} (time to C _{max})	h
AUC (exposure of [plasma] to drug)	ng*h/mL, ng*h ⁻¹ *ml ⁻¹ , uM*h
F (bioavailability)	unitless fraction or %
CL (clearance)	mL/min, L/h, L*h ⁻¹ kg ⁻¹
V _d (volume of distribution)	mL, L
k _{el} (elimination rate constant)	h ⁻¹
k _a (absorption rate constant)	h ⁻¹
T _{1/2} (time required for drug amount to decrease by 50%)	min, h, days